

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (AOC) Examination November 2018

OC-835 Problem Solving in Chemical Sciences I

Date: 23/11/2018 Day: Friday Time: 01.30 pm To 01.50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

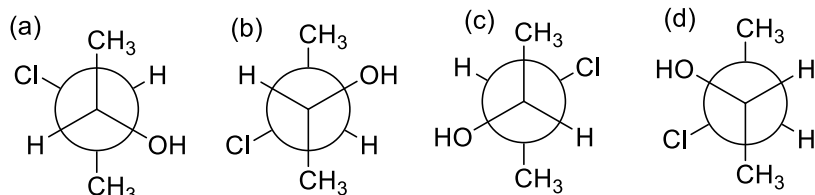
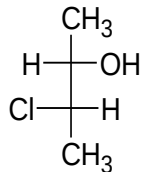
Q - I Choose the correct answer for the following questions. [10]

- 1 What is the order of reaction, if unit of rate constant is $\text{lit}^{1/2} \text{Mol.}^{-1/2} \text{sec}^{-1}$?
(a) 3/2 (b) 1/2
(c) 5/2 (d) zero
- 2 If we plot the graph of $\ln k$ vs. $1/T$ in Arrhenius equation then what is the value of slope?
(a) $-E_a / R$ (b) E_a / R
(c) $\ln A$ (d) E_a
- 3 The packing fraction and number of atomic site per unit cell for BCC is :
(a) 0.52 and 4 (b) 0.68 and 2
(c) 0.68 and 4 (d) 0.52 and 2
- 4 What is the Emf of cell, if standard potential are $E^\circ \text{Fe}^{+2} / \text{Fe} = -0.447 \text{ V}$ and $E^\circ \text{Cd}^{+2} / \text{Cd} = -0.403 \text{ V}$?
(a) -0.85 V (b) 0.044 V
(c) -0.044 V (d) 0.085 V
- 5 The slope and intercept obtained from the graph of $1 / R$ vs. $1 / [S]$ for Michael Menten enzyme catalyzed reaction are 300 and 4×10^5 . What is the value of K_m ?
(a) 4×10^{-5} (b) 7.5×10^{-4}
(c) 1.5×10^{-4} (d) 7.5×10^3

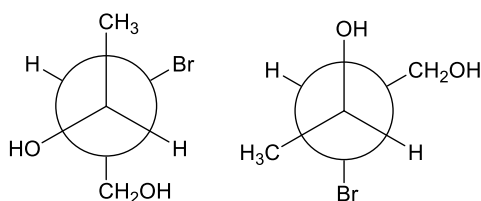
6 The correct relation for half-life time for third order chemical reaction is:

- (a) $1.5 / C_0^2$ (b) $1.5 / C_0^2 k$
 (c) $2 / C_0^2 k$ (d) $1 / C_0^2 k$

7 Enantiomer of the following compound is _____.

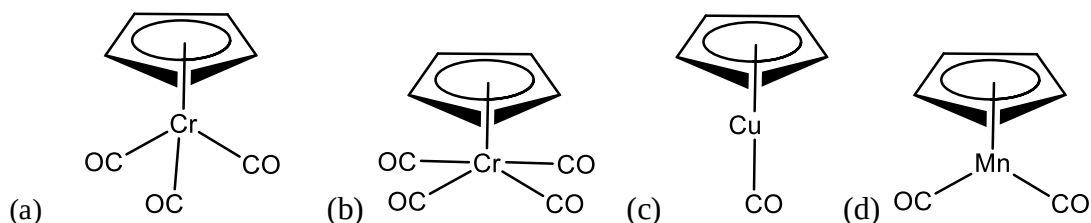


8 The pair of structures given below represent



- (a) Enantiomers (b) Diastereomers (c) Homomers (d) Epimers

9 Identify the specie(s) which follows 18 e⁻ rule.



10 Heating the sample of $[(\eta^5 - C_5H_5)Mo(CO)_3]_2$ results in formation of $[(\eta^5 - C_5H_5)Mo(CO)_2]_2$. The Mo – Mo bond order changes from

- (a) 1 to 3 (b) 1 to 2 (c) 2 to 3 (d) 2 to 4

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OC-835 Problem Solving in Chemical Sciences I

Date: 23/11/2018 Day: Friday Time: 01.50 pm To 03.30 pm Maximum Marks: 25

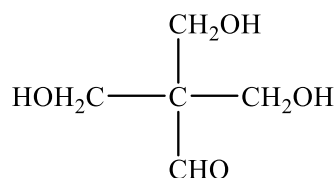
Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed. (Any five) [10]

- 1 The value of resistance is 5 ohm and the value of cell constant is 0.5 cm^{-1} . Calculate the specific conductance of electrolyte solution.
- 2 Find out $\frac{1}{4}$ times life time for 1^{st} order reaction
- 3 Repeated measurement of lead in lake water are 3.2, 5.2, and 7.2 ppm. Calculate standard deviation for Pb.
- 4 Give the synthesis of following compound by Aldol condensation.



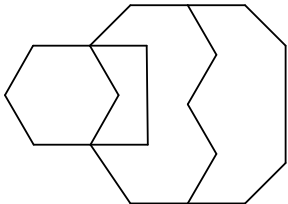
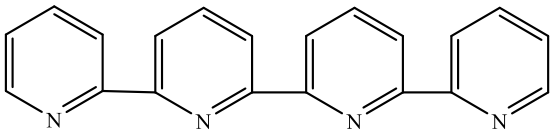
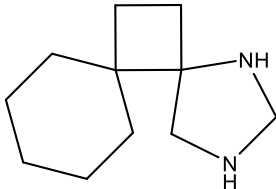
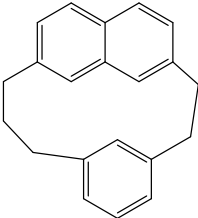
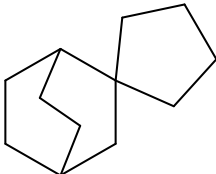
- 5 Find out the hapticity of the following compounds.
 - a) $[(\pi - \text{CP})\text{Ru}(\eta^x - \text{C}_3\text{H}_5)(\text{PPh}_3)]$
 - b) $[(\eta^x - \text{CP})(\eta^y - \text{CP})\text{Co}(\text{CO})_2]$
- 6 Oxidation occurs easily in case of(Show necessary calculations for the correct answer)
 - a) $(\eta^5 - \text{C}_5\text{H}_5)_2\text{Fe}$
 - b) $(\eta^5 - \text{C}_5\text{H}_5)_2\text{Co}$
 - c) $(\eta^5 - \text{C}_5\text{H}_5)_2\text{Ru}$
 - d) $(\eta^5 - \text{C}_5\text{H}_5)_2\text{Co}^+$
- 7 The stable cyclopentadienyl complex of beryllium is
(Justify the answer on the basis of 18 electron rule)
 - a) $[\text{Be}(\eta^2 - \text{C}_5\text{H}_5)_2]$
 - b) $[\text{Be}(\eta^2 - \text{C}_5\text{H}_5)(\eta^3 - \text{C}_5\text{H}_5)]$
 - c) $[\text{Be}(\eta^1 - \text{C}_5\text{H}_5)(\eta^3 - \text{C}_5\text{H}_5)]$
 - d) $[\text{Be}(\eta^1 - \text{C}_5\text{H}_5)(\eta^5 - \text{C}_5\text{H}_5)]$

SECTION – II

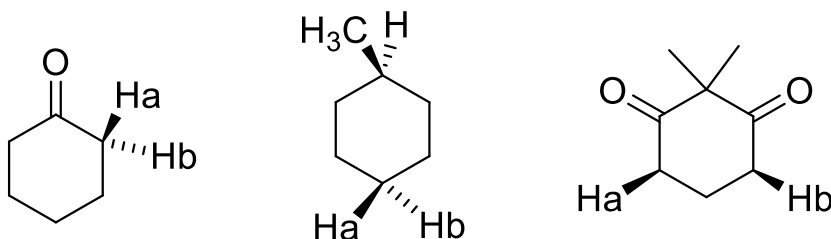
Q-III Answer the following questions as directed. (Any Three)

[15]

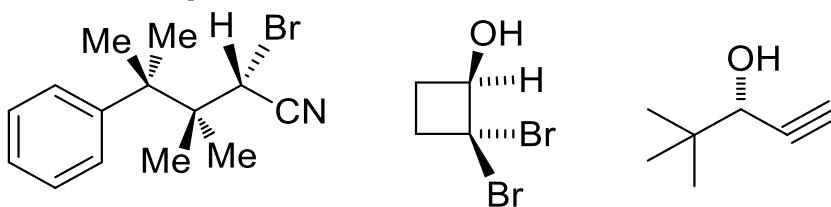
1 State the correct IUPAC name of the following hydrides compounds

i.		iv.	
ii.		v.	
iii.			

2 (i) Explain the topic relationship of H_a with H_b in following compounds using symmetry criteria.



(ii) Assign chirality descriptors to the following compounds and explain your answer on basis of sequence rule.



3 Draw the correct structure of the following hydrides compound names.

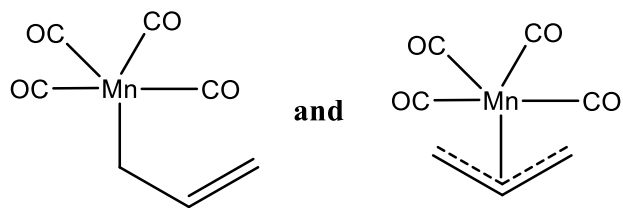
- | | |
|---|-------------------------------|
| (i) Tricyclo [2.2.1.0 ^{2,6}]heptane | (ii) 2-oxa-6-thia-4-azaoctane |
| (iii) Thiobenzoic-o-acid | (iv) Ethyl dithioacetate |
| (v) Thioacetic acetic thioanhydride | |

4 (i) Choose the isoelectronic pair among the following with the necessary calculations.

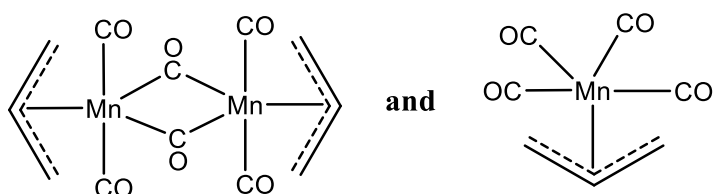
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|--------------------|---------------------------------|
| (a) $[V(CO)_6]$ | (b) $[Cu(\eta^5 - C_5H_5)(CO)]$ |
| (c) $[Co(CO)_4]^-$ | (d) $[Ir(Cl)(CO)(PPh_3)_2]_2$ |

- (ii) The reaction of $\text{Na}[\text{Mn}(\text{CO})_5]$ with allyl chloride gives A along with NaCl . Photolysis of A results in B along with the elimination of CO . The correct structure formulation of compound A and B are (Justify the answer on the basis of 18 electron rule)

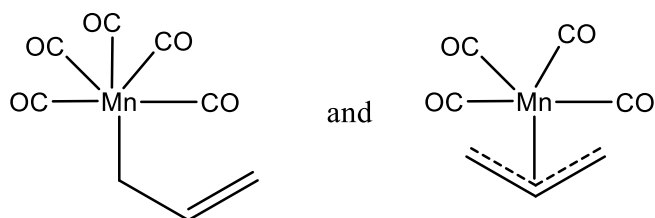
(a)



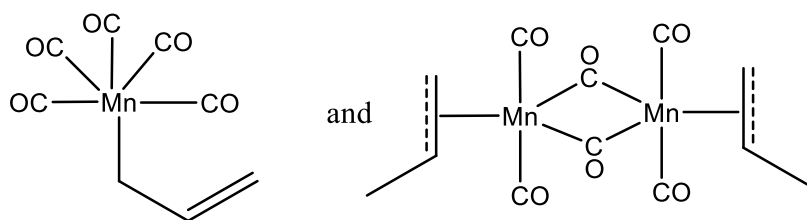
(b)



(c)



(d)



- 5 Write the equation for first order, second order and third order reaction and discuss its characteristics.